Tanzeel Hussain

Full-Stack GenAI Engineer | RAG, AI Agents, MCP | LLM Fine-Tuning (QLoRA) | FastAPI | Flutter

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linkedin.com/in/tanzeelhussain-swe | Portfolio | GitHub | Upwork

Professional Summary

Full-stack GenAI engineer and Software Engineering student at FAST-NUCES who builds retrieval-augmented generation (RAG) systems, AI agents and the apps around them, from the FastAPI backend to the Flutter client. Built CiteRAG, a RAG chatbot whose citations are verified by code, reaching 92% retrieval hit@5 with 681 offline tests. Currently fine-tuning a Qwen2.5 prompt-injection guard with QLoRA and co-building OnCallPilot, an AI incident copilot where a human approves every action. Led the backend and infrastructure for PlayGrid, a top-4 team at NaSCon '26, and completed an 8-week remote Flutter internship at FlutterCraft.app.

Skills

- **LLMs and agents:** RAG, LangChain, LangGraph, LlamaIndex, CrewAI, AI agents and tool calling, Model Context Protocol (MCP), prompt and context engineering, structured output (Pydantic), OpenRouter
- **Retrieval and evaluation:** embeddings, vector databases (Chroma, pgvector), hybrid search (BM25 and vector), re-ranking, semantic caching, RAG evaluation (golden sets, hit@k), prompt-injection defense
- **Fine-tuning:** QLoRA, Qwen2.5, Hugging Face, synthetic data generation, LLM-as-judge filtering, GGUF, Ollama, Google Colab
- **Backend and data:** Python, FastAPI, REST APIs, WebSockets, PostgreSQL, Redis, MQTT, Supabase, Firebase, Streamlit, Node.js, Express
- **Mobile:** Flutter, Dart, Riverpod, go_router, Hive, Drift (SQLite), Firebase Cloud Messaging, offline sync
- **DevOps and cloud:** Docker, Docker Compose, GitHub Actions, Google Cloud Run, Cloudflare Workers
- **Architecture:** Clean Architecture, SOLID, GoF Design Patterns, event-driven design, OOP
- **Languages and tools:** Python, Dart, SQL, C++, Java, Git, GitHub, Postman
- **AI coding tools:** Cursor, Claude Code

Work Experience

Upwork

Remote | Freelance, ongoing

Freelance Flutter Developer | Ongoing

- Build Flutter apps for independent clients, from the screens to the backend they run on
- Integrate Firebase and REST APIs into client apps, delivered remotely alongside a full-time degree

FlutterCraft.app

Remote | Jul 2026 – Aug 2026

Flutter Developer Intern | Jul 2026 – Aug 2026

- Completed an 8-week remote Flutter development internship, from 1 Jul to 28 Aug 2026
- Awarded a Certificate of Internship on 30 Aug 2026 (ID FC-INT-2026-001)

Projects

CiteRAG: Citation-Verified RAG Chatbot | Python, FastAPI, LlamaIndex, Chroma, LangGraph, CrewAI, MCP, Docker | GitHub

- Built a RAG chatbot over PDFs, Markdown and images that verifies every citation in code, fuzzy-matching each quoted passage against its source page and retrying once when a check fails
- Reached 92% retrieval hit@5 on an AI-reviewed golden set with LlamaIndex hybrid search over Chroma and a BM25 docstore, measured by an evaluation harness
- Added 3 answer modes: Standard, Deep (a 3-agent CrewAI crew) and Agent (a LangGraph agent that plans, searches with tools and reflects)
- Served repeat questions in 0.1 s from a semantic cache on NumPy and SQLite, and exposed 3 read-only tools to any MCP client through an MCP server over stdio
- Treated text inside documents as data, never instructions, with 0 of 5 injection canaries leaked; shipped with 681 tests that run without network access

OnCallPilot: Mobile AI Incident Copilot ***(In progress)*** | Python, FastAPI, Flutter, Supabase (Postgres, pgvector), Redis Streams, Docker

- Co-building, in a team of 2, an incident copilot whose agent investigates alerts with read-only tools, ranks up to 3 root causes that cite stored evidence and proposes 1 action from a fixed catalogue
- Own the acting and display stream: the Flutter Android app for evidence review and biometric approval, the isolated runner that executes approved actions, and Chaos Shop, a testbed with 8 fault scenarios
- Designed around a strict safety model: the LLM never touches infrastructure, approvals are signed and single-use, and the runner re-validates every action and can only start, stop or restart allowlisted containers
- To be judged by a 48-run benchmark (manual vs copilot) on time-to-diagnosis with 0 unsafe actions

RU-Guard: Roman Urdu Prompt-Injection Guard ***(In progress)*** | Qwen2.5, QLoRA, FastAPI, Ollama, LangGraph, Google Colab

- Fine-tuning Qwen2.5-0.5B and 1.5B Instruct with QLoRA to detect prompt injection in Roman Urdu, English and mixed text, served locally as a FastAPI guard service on Ollama (GGUF)
- Generated 4,004 training and 496 validation rows with Qwen2.5-14B-Instruct-AWQ on Colab; rejected v1 after a 200-row review found 24% bad rows, then rebuilt it with stricter prompts and an LLM judge
- Froze a 300-row test set behind a SHA-256 lock so it is never trained or tuned on, and placed guard checks at input, retrieval, ingest and output in a reference LangGraph RAG app

PlayGrid: Multi-Sport League Platform | FastAPI, PostgreSQL, Redis, MinIO, MediaMTX, Flutter, Docker | GitHub

- Led the backend and infrastructure in a team of 3, placing top 4 at the NaSCon '26 App Dev Competition
- Built a FastAPI REST and WebSocket API on a PostgreSQL append-only event store, with Redis for rate limiting, event streams and real-time pub/sub, pushing live scores to 25 clients at 41 ms p95
- Added live streaming over RTMP, WebRTC and HLS with MediaMTX, stored receipts and recordings in MinIO, and generated fixtures for 4 tournament formats with 97%+ test coverage across core modules

BioGuard: IoT Cold-Chain Compliance System | FastAPI, Flutter, Riverpod, MQTT, WebSocket, Firebase

- Built an IoT system that monitors temperature-sensitive medicines, with a Python sensor simulator publishing temperature and lock readings over MQTT (QoS 1) to a Mosquitto broker
- Designed a FastAPI backend that checks each reading against per-device thresholds and opens, escalates or resolves alerts, pushed live over WebSocket and Firebase Cloud Messaging
- Built the Flutter app with JWT auth, a live multi-device dashboard, trend charts and PDF compliance reports

AcadAI Buddy: AI Study Companion | Flutter, Riverpod, Firebase, OpenRouter

- Built an AI study app with real-time AI chat, auto-generated MCQ quizzes and note summarization from text or images, covering 40+ subjects including all FAST-NUCES core courses
- Structured it in 4-layer Clean Architecture with Riverpod and dartz Either error handling on Firebase

CreateResume AI: AI Resume Builder | Flutter, Supabase, Riverpod, OpenRouter

- Built an AI resume builder that turns plain-language input into ATS-optimized resumes, exported as PDFs across 5 templates
- Kept the OpenRouter API key out of the app by generating resumes in a Supabase Edge Function, with credit-based free and premium plans

FounderAI: AI Startup Assistant | Node.js, Express, Render

- Built an AI startup assistant at the Google Developer Groups "Build with AI" hackathon

Computer science fundamentals

Console Social Media Platform | C++, Data Structures

- Built without STL containers in a team of 2: a hash table for users, a graph with BFS and DFS for friends, linked lists for feeds, and an AVL tree for top-K analytics

UniDisc: University Management System | C++, Java, Discrete Mathematics

- Modelled course scheduling, prerequisites and enrollment with predicate logic, set and relation analysis, and dynamic programming for conflict detection

Education

B.S. Software Engineering | FAST-NUCES | Faisalabad, Pakistan | 2024 – 2028 (expected)

Coursework: Data Structures, Object-Oriented Programming, Design Patterns, Software Design and Architecture